Beyond their role in fashioning the laws governing nature's forces, ideas of symmetry are vital to the concept of time itself. No one has as yet found the definitive, fundamental definition of time, but, undoubtedly, part of time's role in the makeup of the cosmos is that it is the bookkeeper of change. We recognize that time has elapsed by noticing that things now are different from how they were then. The hour hand on your watch points to a different number, the sun is in a different position in the sky, the pages in your unbound copy of *War and Peace* are more disordered, the carbon dioxide gas that rushed from your bottle of Coke is more spread out—all this makes plain that things have changed, and time is what provides the potential for such change to be realized. To paraphrase John Wheeler, time is nature's way of keeping everything—all change, that is—from happening all at once.

The existence of time thus relies on the *absence* of a particular symmetry: things in the universe must *change* from moment to moment for us even to define a notion of *moment to moment* that bears any resemblance to our intuitive conception. If there were perfect symmetry between how things are now and how they were then, if the change from moment to moment were of no more consequence than the change from rotating a cue ball, time as we normally conceive it wouldn't exist.[3] That's not to say the spacetime expanse, schematically illustrated in Figure 5.1, wouldn't exist; it could. But since everything would be completely uniform along the time axis, there'd be no sense in which the universe evolves or changes. Time would be an abstract feature of this reality's arena—the fourth dimension of the spacetime continuum—but otherwise, it would be unrecognizable.

Nevertheless, even though the existence of time coincides with the lack of one particular symmetry, its application on a cosmic scale requires the universe to be highly respectful of a different symmetry. The idea is simple and answers a question that may have occurred to you while reading Chapter 3. If relativity teaches us that the passage of time depends on how fast you move and on the gravitational field in which you happen to be immersed, what does it mean when astronomers and physicists speak of the entire universe's being a particular definite age—an age which these days is taken to be about 14 billion years? Fourteen billion years according to whom? Fourteen billion years on which clock? Would beings living in the distant Tadpole galaxy also conclude that the